A reproducible spectral certificate for the nonexistence of homogeneous quartic rotation symmetric bent functions in twelve variables

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Abstract

We give a computer-assisted certificate excluding bent functions from the 43-dimensional space of homogeneous quartic rotation symmetric Boolean functions in twelve variables. The argument uses the restrictions $g_z(u) = f(u, u+z)$, necessary energy identities at all 64 fiber frequencies, and finite Poisson constraints. A triangular parametrization covers all 2^{43} coefficient vectors without evaluating each complete truth table. The final two stages retain 128 classes and then none. Two implementations, one using signed spectral transport and one evaluating each fiber directly, reproduce the same stage counts. We state the necessity of every family of filters and distinguish exact computation from the auxiliary statements checked in Lean. The result concerns twelve variables; it does not settle the conjecture in arbitrary dimension.

1 Scope and conventions

The nonexistence conjecture for homogeneous rotation symmetric bent functions of degree greater than two is discussed in [1, 2, 3]. This paper addresses the quartic case in twelve variables with a reproducible computational certificate. We do not claim priority over all earlier searches or claim a new general obstruction for every even dimension. The mathematical and executable descriptions below are intended to make the finite result independently assessable.

Write R_n for left rotation on n bits. Integers encode vectors with coordinate zero in the least significant bit, and addition means XOR. For a Boolean function $f : \mathbb{F}_2^n \to \mathbb{F}_2$, put

$$W_f(a) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x)+\langle a,x\rangle}.$$

For even n, bentness means $|W_f(a)| = 2^{n/2}$ for every a. In this paper $n = 12$, $u, z, b \in \mathbb{F}_2^6$, and

$$g_z(u) = f(u, u+z), \quad W_z(b) = W_{g_z}(b), \quad A_z = W_z(0).$$

Let $\mathcal{H}$ be the binary span of the rotation orbits of squarefree degree-four monomials. Orbits are ordered by the lexicographically smallest support tuple. There are 43 orbits: forty of size 12, two of size 6, and one of size 3. Thus $\dim \mathcal{H} = 43$ and $|\mathcal{H}| = 2^{43}$. The zero function is included and is harmless for the exclusion statement.

Theorem 1. *No function in $\mathcal{H}$ is bent.*

The proof consists of the necessary conditions in Sections 2–4, exhaustive enumeration in the triangular coordinates of Section 5, and the execution records supplied with the source. This is a computer-assisted theorem, not an end-to-end Lean formalization.

2 Fiber identities and signed transport

For any $\alpha, \beta \in \mathbb{F}_2^6$, changing variables from (x, y) to $(u, z) = (x, x + y)$ gives

$$W_f(\alpha, \beta) = \sum_z (-1)^{\langle \beta, z \rangle} W_z(\alpha + \beta). \tag{1}$$

Consequently, if f is bent, Parseval in the variable z gives

$$\sum_z W_z(b)^2 = 4096 \qquad (b \in \mathbb{F}_2^6). \tag{2}$$

For any set S of fibers, the partial energy $E_S(b) = \sum_{z \in S} W_z(b)^2$ is therefore at most 4096. These 64 identities are necessary for bentness; they are not a substitute for testing all 4096 coefficients of W_f.

Rotation by six positions interchanges the two blocks, so $g_z(u + z) = g_z(u)$. Pairing u with $u + z$ proves

$$\langle b, z \rangle = 1 \quad \Longrightarrow \quad W_z(b) = 0. \tag{3}$$

For $0 \le k < 6$, define $\tau_k(z)$ as the low six bits of $R_{12}^k(0, z)$. Rotation sends $(u, u + z)$ to $(R_6^k u + \tau_k(z), R_6^k u + \tau_k(z) + R_6^k z)$. Hence

$$W_{R_6^k z}(b) = (-1)^{\langle b, \tau_k(z) \rangle} W_z(R_6^{-k} b). \tag{4}$$

The sign cannot be omitted. Squared energies do not detect its omission, but the Poisson sums below do. In particular, A_z is constant on each cyclic orbit of z.

3 Finite algebraic foundations

All finite checks in this section are performed independently by `audit_foundations.py`; their explicit orbit table and values are in `results/foundations.json`. These checks use only the 43 generators or small restriction spaces, not a random sample of $\mathcal{H}$.

Lemma 2. *For every $f \in \mathcal{H}$, $8 \mid W_z(b)$ for all z, b. Moreover, $16 \mid A_z$ for $z = 9, 27, 63$, $W_9(63) = W_{63}(63) = 0$, and $16 \mid W_{27}(63)$.*

Proof. The algebraic normal form of every generator restricted to every fiber has degree at most three, as verified by the binary Möbius transform. Degree at most three is preserved by binary linear combinations. For $z \neq 0$, translation invariance makes g_z a function on the five-dimensional quotient by $\langle z \rangle$. If $\langle b, z \rangle = 0$, the same holds for $g_z + \langle b, u \rangle$. A polynomial of degree less than five on $\mathbb{F}_2^5$ has even weight: summing each monomial over the cube gives zero modulo two. Its Walsh sum is divisible by four, and lifting to $\mathbb{F}_2^6$ doubles this sum. Otherwise use (3).

At $z = 0$, only the three short monomial orbits survive; their restrictions are Q_1, Q_2, Q_3, where

$$Q_d = \sum_{i=0}^{5} u_i u_{i+d} \ (d = 1, 2), \qquad Q_3 = u_0 u_3 + u_1 u_4 + u_2 u_5.$$

The eight combinations have all their Walsh coefficients divisible by eight. This is a finite check on $8 \cdot 64$ coefficients.

For $z = 9, 27$, the induced affine map is $T_z(u) = R_6^3 u + \tau_3(z)$, with *correct* constants $\tau_3(9) = 1$ and $\tau_3(27) = 3$. It satisfies $T_z^2(u) = u + z$, so its sixteen orbits all have size four. The value of g_z is constant on each orbit. Each of the 43 generators is one on an even number of these orbits. This parity condition is linear in the generator values, so it holds for every combination. Thus $8 \mid \mathrm{wt}(g_z)$ and $16 \mid A_z$. For $z = 9$, T_z reverses the parity of u, giving

$W_9(63) = 0$. For $z = 27$, that parity is constant on each orbit and equals one on eight of the sixteen orbits; the same even-weight argument gives $16 \mid W_{27}(63)$.

For $z = 63$, the negacyclic shift $u \mapsto R_6 u + 1$ has five orbits of length 12 and one of length 4, represented by $0, 2, 4, 6, 10, 12$. Let their values be $v_1, \ldots, v_6$. Direct evaluation on each of the 43 generators gives $v_1 + \cdots + v_6 = 0$ in $\mathbb{F}_2$; linearity extends this to every f. If $k = \sum_{i=1}^{5} v_i$ and $\delta = v_6$ as integers, $k \equiv \delta \pmod 2$ and

$$\mathrm{wt}(g_{63}) = 12k + 4\delta = 24q + 16\delta, \qquad A_{63} \in \{-64, -32, -16, 16, 32, 64\}.$$

The negacyclic shift reverses parity, so $W_{63}(63) = 0$. □

For redundancy, enumerating the complete images of the three restriction maps gives ranks $9, 10, 5$ for $z = 9, 27, 63$, respectively, and confirms the assertions on all $512 + 1024 + 32$ image functions.

Proposition 3. *If $f \in \mathcal{H}$ is bent, $16 \mid A_0$ and $c_{40} + c_{26}c_{42} = 1$ in $\mathbb{F}_2$, where indices are zero-based.*

Proof. The nonzero fiber orbits have representatives $1, 3, 5, 7, 9, 11, 13, 15, 21, 23, 27, 31, 63$. Only $9, 27, 63$ have odd orbit sizes. Divide (2) at $b = 0$ by 64 and reduce modulo two; Lemma 2 leaves $(A_0/8)^2 = 0$ modulo two. At $b = 63$, all odd-weight fibers vanish. The even-weight fibers 9 and 63 also vanish. The only other nonzero orbits have representatives $3, 5, 15, 23, 27$; their total energies are divisible by 384 (size six times 8^2, or size three times 16^2). Thus $W_0(63)^2 \equiv 4096 \equiv 256 \pmod{384}$. The eight values for (c_{26}, c_{40}, c_{42}) in lexicographic order are

(c_{26}, c_{40}, c_{42})	000	001	010	011	100	101	110	111
A_0	64	8	0	16	16	32	0	−8
$W_0(63)$	0	−8	16	−16	0	−32	−32	8

and give the stated equivalence. Exactly four sectors remain. □

4 Necessity of all sieve filters

4.1 Subspace Poisson constraints

For any binary subspace $V \leq \mathbb{F}_2^6$, any $b, \gamma \in \mathbb{F}_2^6$, define $L_V(b, \gamma) = \sum_{z \in V} (-1)^{\langle \gamma, z \rangle} W_z(b)$. Fourier inversion of (1) gives

$$L_V(b, \gamma) = \frac{|V|}{64} \sum_{\beta \in \gamma + V^\perp} W_f(b + \beta, \beta). \tag{5}$$

For bent f, each summand is 64 times a sign. Rotation symmetry forces identical signs on each cyclic orbit of the 12-bit frequency. If the coset has orbit multiplicities $m_1, \ldots, m_t$, necessarily

$$L_V(b, \gamma) \in |V| \left\{ \sum_{j=1}^{t} m_j \epsilon_j : \epsilon_j \in \{-1, 1\} \right\}. \tag{6}$$

Both programs construct this finite set by repeated signed addition. Allowing independent signs in separate tests enlarges the feasible set; it cannot discard a bent function. The code checks closure under XOR for each subspace before constructing its table.

The 24 spaces used are the thirteen lines $\langle r \rangle$ for the nonzero representatives listed above, nine further spaces in the table, and the final two spaces V_3' and V_4.

Code name	Elements or basis (span over $\mathbb{F}_2$)
`V2_21`	$\{0, 21, 42, 63\}$
`V2_27`	$\{0, 27, 45, 54\}$
`V2_9a`, `V2_9b`	$\langle 9, 18\rangle$, $\langle 9, 54\rangle$
`V2_7a`, `V2_7b`	$\langle 7, 56\rangle$, $\langle 7, 14\rangle$
`V2_7c`, `V2_7d`	$\langle 7, 21\rangle$, $\langle 7, 28\rangle$
`V3`	$\langle 9, 18, 36\rangle$
V_3' (`V3_7a`)	$\langle 63, 9, 7\rangle$
V_4 (`V4`)	$\langle 21, 42, 9, 18\rangle$

In particular, $V_2 = \langle 21, 42\rangle$ and $V_3 = \langle 9, 18, 36\rangle$ are not nested: $V_2 \cap V_3 = \langle 63\rangle$ and $V_2 + V_3 = V_4$.

4.2 Energy congruences

For each cyclic set S of fibers processed by the sieve, the following congruences are used only at the indicated stages.

1. After the initial fibers $0, 63, 21, 42$, the residual energy at $b = 0$ is divisible by 384. The unprocessed size-six orbits contribute multiples of $6 \cdot 64$, and the size-three orbits contribute multiples of $3 \cdot 256$ by Lemma 2.

2. Once $9, 18, 36, 27, 54, 45$ have also been processed, every remaining fiber orbit has length six. Residual energies at $b = 0, 63$ are divisible by 384, since these frequencies are fixed by rotation and squares in (4) agree. The initial $b = 63$ test is the stronger sector congruence in Proposition 3.

3. At $b = 21$, $R_6^2 b = b$. Outside $\{0, 21, 42, 63\}$, the R_6^2-orbits have size three. Therefore residual energy is divisible by $3 \cdot 64 = 192$ at every stage after the initial one.

4. After processing the size-three fiber orbits, residual energy summed over the three frequencies $\{9, 18, 36\}$ or $\{27, 54, 45\}$ is divisible by 192. Indeed rotation acts on the remaining pairs (z, b) in length-six orbits, and each squared term is divisible by 64; this even gives divisibility by 384.

The size-three contribution at $b = 21$ can be computed as $3W_{27}(21)^2$. Indeed, $R_6^{-2}21 = 21$, so two successive rotations preserve the squared coefficient in (4). On a three-element fiber orbit, this implies equality of all three squares. This explains the prefilter in the direct implementation.

4.3 Autocorrelation and diagonal Walsh congruences

Equation (1) at $b = 0$ says that the Fourier transform of A consists of the diagonal coefficients $W_f(\beta, \beta)$. If f is bent, inversion gives $\sum_z A_z A_{z+63} = 0$. Dividing by two and grouping complementary orbits yields the initial necessary condition

$$A_0 A_{63} + A_{21}^2 \equiv 0 \pmod 3.$$

Once the size-three orbits are known it yields

$$A_0 A_{63} + A_{21}^2 + 3A_9 A_{27} \equiv 0 \pmod{192}. \tag{7}$$

For distinct complementary size-six orbits the residual contribution is $6A_r A_s$; for a self-complementary orbit it is $3A_r^2$. Both are divisible by 192.

The diagonal Walsh coefficients at $\beta = 0, 63$ are ± 64. All omitted terms in their cyclic orbit sums are divisible by 48: size-six orbits have $8 \mid A_r$, and size-three orbits have $16 \mid A_r$. Hence the initial tests are

$$A_0 + A_{63} \pm 2A_{21} \equiv 16 \text{ or } 32 \pmod{48}. \tag{8}$$

These tests use $\beta = 0, 63$; $b = 21$ in the energy tests is a different frequency variable.

5 Exhaustive triangular enumeration

Let B be the 43×4096 matrix whose rows are generator truth tables, with columns ordered by (z, u), grouping complete fiber orbits as in Table 1. Row reduction over $\mathbb{F}_2$ gives $M = PB$, with P invertible and 43 pivots. The direct implementation explicitly checks $PB = M$ and that the pivot submatrix of M is the identity. Thus it also checks injectivity of the coefficient map.

Group rows by the stage containing their pivot. A later group is zero on all earlier fiber columns. Enumerate all binary combinations of the first group, filter, and extend each survivor by all combinations of the next group. Every original coefficient vector has exactly one such sequence of extensions. No future choice can change a quantity already tested, and every test is necessary for bentness. Therefore a zero terminal count proves Theorem 1.

Table 1: Exact counts. Counts refer to classes of restrictions at the stated rank, not equally sized lists of complete functions.

New fiber orbit(s)	Rank	Rank increment	Survivors
$0, 63, 21, 42$: initial congruences	12	12	712
Same fibers: energy and Poisson	12	0	361
$\mathcal{O}_9$	18	6	20,158
$\mathcal{O}_{27}$	24	6	212,296
$\mathcal{O}_7$	27	3	8,688
$\mathcal{O}_1$	31	4	26,496
$\mathcal{O}_{11}$	34	3	94,936
$\mathcal{O}_{13}$	36	2	149,448
$\mathcal{O}_{31}$	37	1	106,552
$\mathcal{O}_3$	41	4	349,680
$\mathcal{O}_5$	41	0	23,636
$\mathcal{O}_{15}$	42	1	128
$\mathcal{O}_{23}$	43	1	0

At rank 12 the initial conditions are the energy bounds at $b = 0, 21$, their residual congruences, the sector congruence at $b = 63$, the autocorrelation condition modulo three, and (8). The next filter adds all 64 energy bounds and Poisson on $\langle 63 \rangle, \langle 21 \rangle, V_2$. At rank 18 add all energy bounds and Poisson on $\langle 9 \rangle$. At rank 24 add the energy congruences, (7), the aggregate energy congruences, and Poisson on the spaces named `V2_27`, `V1_27`, `V2_9a`, `V2_9b`, `V3`. At rank 27 add the seven spaces involving 7 or V_4 in Section 4. Thereafter, each new orbit adds its energy and Poisson on its representative line. Residual congruences at $0, 21$ remain active; the 63 congruence is also applied after representatives $3, 5, 15$. At the last stage, all 64 energies must equal 4096. Every Poisson check covers all 64 frequencies and coset characters; early checks on subsets merely reject failures sooner.

The rank-41 count after orbit 5 represents $23,636 \cdot 4 = 94,544$ full coefficient vectors. Expanding these without the remaining Poisson tests gives 32,176 satisfying energy at $b = 0$,

39,520 at $b = 63$, 5,552 at both, and zero at all 64 values of b. This redundancy again uses necessary energy conditions, not an asserted equivalence with bentness.

6 Reproduction and trust boundary

The release contains three Python programs and the execution logs. The main sieve uses 14 representative fibers and the signed transport formula. The second sieve uses coefficient vectors in $\mathbb{F}_2^{43}$, all 64 fiber truth tables, and a butterfly Walsh transform; it never uses signed transport to obtain a fiber spectrum. They share the mathematical filters, so agreement addresses implementation errors but is not an independent proof that those filters are necessary. Section 4 supplies that proof.

All arithmetic relevant to acceptance is exact. Walsh values on a fiber lie in $[-64, 64]$, total fiber energy is at most 262,144, and all such quantities fit the signed 32-bit arrays used by the sieves. The binary matrix products in unsigned bytes are subsequently reduced modulo two; reduction modulo 256 preserves their parity. The foundation checks use signed 64-bit transforms.

The main program tests sixteen quadratic RS bent functions against the 24 Poisson tables, and checks transport and Poisson identities against complete Walsh transforms on these and fifty deterministic random quartic examples. The quadratic controls are not asserted to pass quartic-specific sector conditions. The foundation script checks signed transport on all 43 generators, all fibers and frequencies, as well as complete Walsh exclusion for $n = 4, 6, 8$.

The Lean file proves finite arithmetic identities, the corrected quarter-period orbit assertion, and the parity syzygy evaluated on the actual list of 43 monomial orbit representatives. Its compilation does not formalize completeness of that list, the whole Fourier argument, the Python execution, or Theorem 1. The release records Lean 4.15.0 and the Python/NumPy versions.

7 Relation to smaller cases and earlier work

The case $n = 4$ is elementary: the only nonzero homogeneous quartic function is $x_0x_1x_2x_3$, with $W_f(0) = 14$, so it is not bent. It is a useful convention check, not evidence of a new general theorem. The supplied broader verifier also reproduces the $n = 10$ correction: the residues of ± 8 and ± 32 modulo five form the same set. Its nonzero-frequency derivative checks, not the invalid sign argument, exclude the remaining branch. These computations are supplementary and are not required for Theorem 1.

Sun et al. [3], Theorem 2, give necessary orbit conditions under explicit prime-size hypotheses. Their condition for $n = 2p^r$, $p \geq 5$, already covers the odd number of short orbits condition in degree four: a stabilizer of a four-element support acts freely on that support, so its order divides both four and $2p^r$. Thus the only short quartic orbits have length p^r, and are bi-antipodal. For $n = 12$ their stated hypothesis with $v_2(n) = 2$ would require the odd prime factor to exceed 16, and therefore does not apply directly. This comparison with one theorem is not a claim that no other published result covers twelve variables. The contribution presented here is the explicit certificate and its auditable spectral reduction.

Data and code. Source files, exact output records, a checksum manifest, and reproduction instructions accompany this manuscript. No external dataset is needed.

Limitations. The certificate does not settle any dimension $n \geq 14$. Tests on generators in larger dimensions establish only the specific linear identities checked; they are not exhaustive bentness searches over the full coefficient spaces.

References

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